

4. **Why are dewdrops not formed on a cloudy night?**

- (a) Clouds absorb the radiation released from the Earth's surface.
- (b) Clouds reflect back the Earth's radiation.
- (c) The Earth's surface would have the low temperature on cloudy nights.
- (d) Clouds deflect the blowing wind to ground level.

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Ans. (b)

Dew : Dew is the water droplets formed by the condensation of water vapour on a relatively cold surface of an object. It forms when the temperature of an object drops below the dew point temperature.

When there is a cloudy weather conditions, terrestrial radiation is radiated back to the earth's surface after reflection from clouds. This leads to the formation of greenhouse conditions due to which temperature on the earth's surface is relatively higher. Hence condition becomes unfavorable for the creation of dew.

The Atmospheric Pressure

*Air occupies space and has weight. The atmosphere exerts vertical pressure on the earth's surface. This is called **atmospheric pressure**.

* Atmospheric pressure is measured using a barometer. It is measured, as the force exerted per unit area of the barometer. A **sudden fall in the level of barometer** is a sign of an upcoming storm. If the readings of a barometer fall and then start rising slowly it denotes that there will be rainfall.

* A continuously **rise in the level of mercury** in the barometer is a sign of an anticyclone and clear weather.

* The horizontal distribution of atmospheric pressure is shown through an isobar. * An **isobar** is an imaginary line that connects points having same atmospheric pressure at mean sea level.

*Based on the process of origin, pressure belts can be divided into two parts-

(1) Based on thermogenesis – pressure belts thus formed are equatorial low-pressure belts and polar high-pressure belts.

(2) Based on motion-pressure belts thus formed are sub-tropical high-pressure belts and sub-polar low-pressure belts.

***Equatorial Low-Pressure Belts** – The belt extends between 5° north and south of Equator. The equator receives vertical rays of the sun throughout the year. As a result, the air gets heated and starts rising up. Hence the region always experiences low pressure. The surface air of the region moves

with very slow velocity as a result of which the environment here is quite calm. This is the reason this pressure belt is also called **Doldrums**.

• **Sub-tropical High-Pressure Belt** – the belt is found between 30°-35° latitudes in both hemispheres. The air rising from the equator starts becoming cooler in the upper layers of atmosphere and hence begins sinking near 30°-35° latitudes. Simultaneously the air from the sub-polar low-pressure belts also starts descending in this region due to Earth's rotation thus a high-pressure zone is created which gives birth to sub-tropical high pressure belt. The pressure belt is highly mobile in nature, thus it is also called **Horse Latitude**.

• **Sub-Polar Low-Pressure Belt** – This pressure belt is found between 60°-65° latitudes in both hemispheres. Though the temperature in this region is low throughout the year the region experiences low pressure. This is due to the fact that rotation of the Earth causes the air to spread and displaces it. As a result low pressure is created.

• **Polar High-Pressure Belts** – The poles and the regions in close proximity to them experience high pressure because the region is very cold which causes the air to become heavy and start descending creating high pressure.

1. **Assertion (A) : 60°–65° latitudes in both the hemispheres have a low-pressure belt instead of high pressure.**

Reason (R) : The low-pressure areas are stable over oceans rather than on land.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

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Ans. (b)

The sub-polar low-pressure belt lies between 60°- 65° latitude in both hemispheres throughout the year as there are low temperatures and low atmospheric pressure. Although this belt has nothing to do with temperature, assertion (A) is correct.

The continuity of sub-polar low pressure belt is disrupted in the Northern Hemisphere due to large landmass along 60°N latitude while in Southern hemisphere this low pressure belt is more pronounced and stable due to the vast ocean presence and lack of landmass.